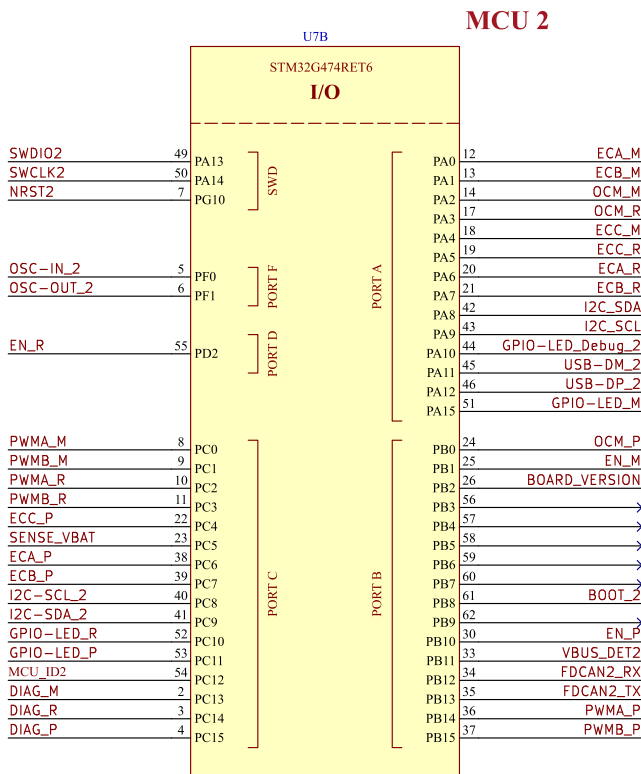
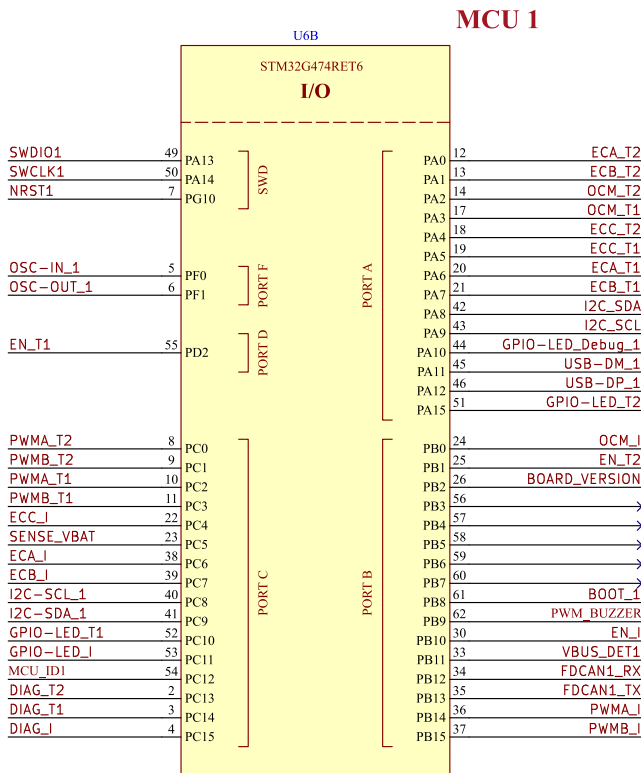


[7] MCU - IOs



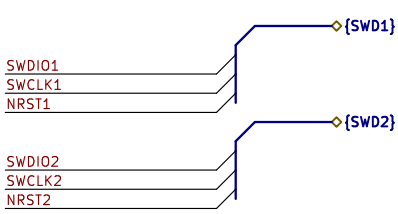
DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

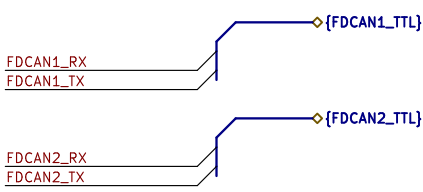
LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

SENSE_VBAT → SENSE_VBAT

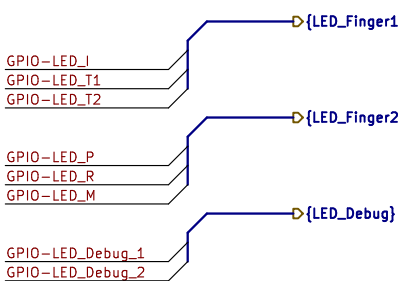
Battery Voltage Sense



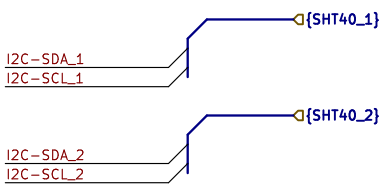
SWD (Debug) Port



5Mbps FD-CAN Bus



LEDs



Temperature and Humidity



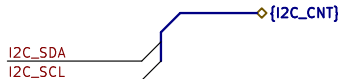
USB PORT

BOARD_VERSION → BOARD_VERSION

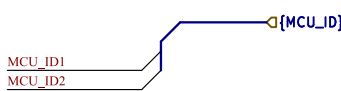
Board Version Selection

PWM_BUZZER → PWM_BUZZER

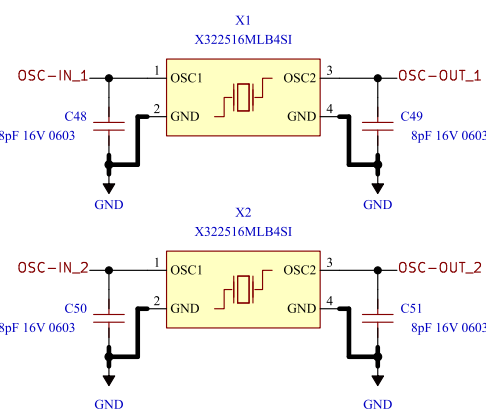
Buzzer



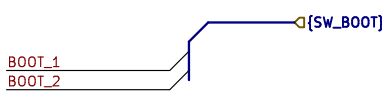
MCB to MB - I2C



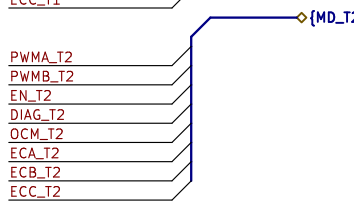
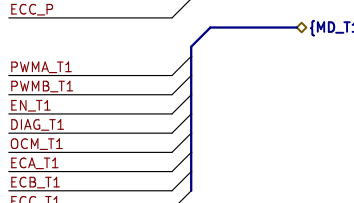
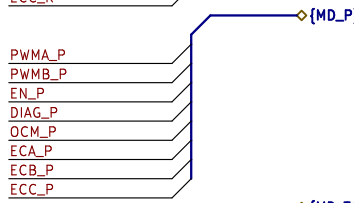
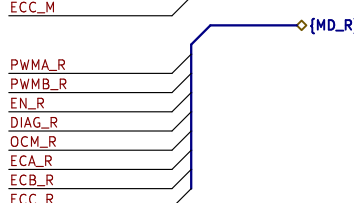
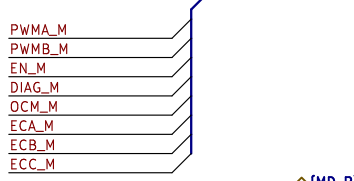
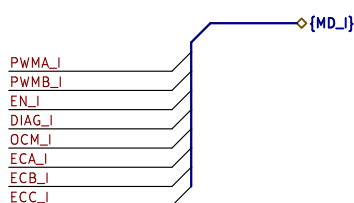
MCU ID



Oscillator HSE



SWITCH



Motor Driver

Battery Voltage Sense

SENSE_VBAT → TP9

Board Version Selection

BOARD_VERSION → TP13

SWD (Debug) Port

NRST2 → TP17
SWCLK2 → TP19
SWDIO2 → TP21

5Mbps FD-CAN Bus

FDCAN1_TX → TP29
FDCAN1_RX → TP31
FDCAN2_TX → TP34
FDCAN2_RX → TP36

LEDs

GPIO-LED_I → TP38
GPIO-LED_T1 → TP40
GPIO-LED_T2 → TP42
GPIO-LED_P → TP44
GPIO-LED_R → TP46
GPIO-LED_M → TP48
GPIO-LED_Debug_1 → TP50
GPIO-LED_Debug_2 → TP52

Temperature and Humidity

I2C-SDA_1 → TP54
I2C-SCL_1 → TP56
I2C-SDA_2 → TP58
I2C-SCL_2 → TP60

USB PORT

USB-DP_1 → TP64
USB-DM_1 → TP66
VBUS_DET1 → TP67

USB-DP_2 → TP69
USB-DM_2 → TP71
VBUS_DET2 → TP73

MCB to MB - I2C

I2C_SCL → TP77
I2C_SDA → TP79

Buzzer

PWM_BUZZER → TP81

MCU ID

MCU_ID1 → TP86
MCU_ID2 → TP88

MD Index

PWMA_I → TP10
PWMB_I → TP11
EN_I → TP12
DIAG_I → TP14
OCM_I → TP15
ECA_I → TP16
ECB_I → TP18
ECC_I → TP20

MD Middle

PWMA_M → TP23
PWMB_M → TP25
EN_M → TP27
DIAG_M → TP28
OCM_M → TP30
ECA_M → TP32
ECB_M → TP33
ECC_M → TP35

MD Ring

PWMA_R → TP37
PWMB_R → TP39
EN_R → TP41
DIAG_R → TP43
OCM_R → TP45
ECA_R → TP47
ECB_R → TP49
ECC_R → TP51

MD Pinky

PWMA_P → TP53
PWMB_P → TP55
EN_P → TP57
DIAG_P → TP59
OCM_P → TP61
ECA_P → TP62
ECB_P → TP63
ECC_P → TP65

MD Thumb 1

PWMA_T1 → TP68
PWMB_T1 → TP70
EN_T1 → TP72
DIAG_T1 → TP74
OCM_T1 → TP75
ECA_T1 → TP76
ECB_T1 → TP78
ECC_T1 → TP80

MD Thumb 2

PWMA_T2 → TP82
PWMB_T2 → TP83
EN_T2 → TP84
DIAG_T2 → TP85
OCM_T2 → TP87
ECA_T2 → TP89
ECB_T2 → TP90
ECC_T2 → TP91

Test Points

Comments:	Company: Willas Willas Electronic Corp.	Variant: DRAFT	Git Hash: 538a591
Sheet Title:	Board Name: Motor Control Board	Project Name: Master Hand	
Sheet Path: /Project Architecture/MCU - IOs/	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27
		Reviewer:	Revision: + (Unreleased)
		Size: A3	Sheet: 7 of 23